① Findout all the list of companies which are conducting placement drive using DSA. (10 companies)

① Amazon
② Google
③ Microsoft
④ Meta
⑤ Uber
⑥ Apple
⑦ Flipkart
⑧ Adobe
⑨ TCS
⑩ Infosys
⑪ Wipro
⑫ Capgemini
⑬ Oracle
⑭ Cognizant
~~⑮~~

② What is the syllabus of DSA in GATE?

Key topics to focus on DSA for GATE – CS Exam

1. Arrays and strings
2. Linked lists
3. Stacks & Queues
4. Trees & Graphs
5. Sorting and Searching algorithms
6. Dynamic programming
7. Hashing
8. Graph Algorithms
9. Heap
10. Divide & Conquer
11. Greedy Algorithms
12. Time & Space complexity.
13. Bit manipulation
14. Trie
15. Segment Tree
16. Graph Traversal Technique

(3) Findout what are the set of variables in
Amazon Invoice Bill
GasBill → variables
Current bill → variables

Amazon

int quantity
float price
Product Name → char/string
Address → string
double discount
Order ID - string
Date - integer
Buyer name - string

Gas Bill

Bill number - string
Consumer number - string
LPG ID - string
Booking reference number - string
Consumername - string
Gas agency name - string
Gas company - Categoricalstring
No. of cylinders - int

Currentbill

Invoice Number/ Bill number - string

Customer name - string

Customer address - string

Date - int

Unitprice - Float

Total amount - Float

GST - float

Meter number - string

Mobile number - string

HW

Write a C++ code which includes a class student. Findout the space occupied by the object.

~~IIIrd~~

Write a C++ code which includes a class Student. Findout the space occupied by the Object.

```
#include<iostream>
using namespace std;
class Student
{
Private:
int rollNo;
char name[30];
float marks;
Public:
void getData()
{
cout<<"Enter RollNumber:";
cin>>rollNo;
cout<<"Enter Name:";
cin>>name;
cout<<"Enter Marks:";
cin>>marks;
}
void displayData()
{
cout<<"\n Roll Number:"<<rollNo<<endl;
cout<<"Name:"<<name<<endl;
cout<<"Marks:"<<marks<<endl;
}
};
int main()
{
Student S1;
S1.getData();
S1.displayData();
cout<<"\n Space occupied by Student Object:"
    <<sizeof(S1)<<"bytes"<<endl;
return 0;
}
```

Sample Output:

Enter Roll Number: 101
Enter Name: Triveni
Enter Marks: 95.5
Space Occupied by Student Object: 40bytes

Given an integer array and integer target value. find the indices of two numbers in the array whose sum is equal to given target value

Given two arrays which are unsorted. Apply bubble sort on these two arrays & merge them into single array. Make sure to eliminate duplicate values.